

THE SKELETAL SYSTEM

General Anatomy Laboratory | Study Guide

OBJECTIVES

- Know how to differentiate between bone and cartilage.
- Know the divisions of the skeleton.
- Know the types of bone.
- Describe the structure of bone.

PLAN

1. Introduction & Definitions
2. Structure & Classification
3. Morphology & Histology
4. Internal & External Structure
5. Development, Growth & Pathology

1. INTRODUCTION & DEFINITIONS

INTRODUCTION

The human skeleton is the bony framework that supports the body and protects internal organs.

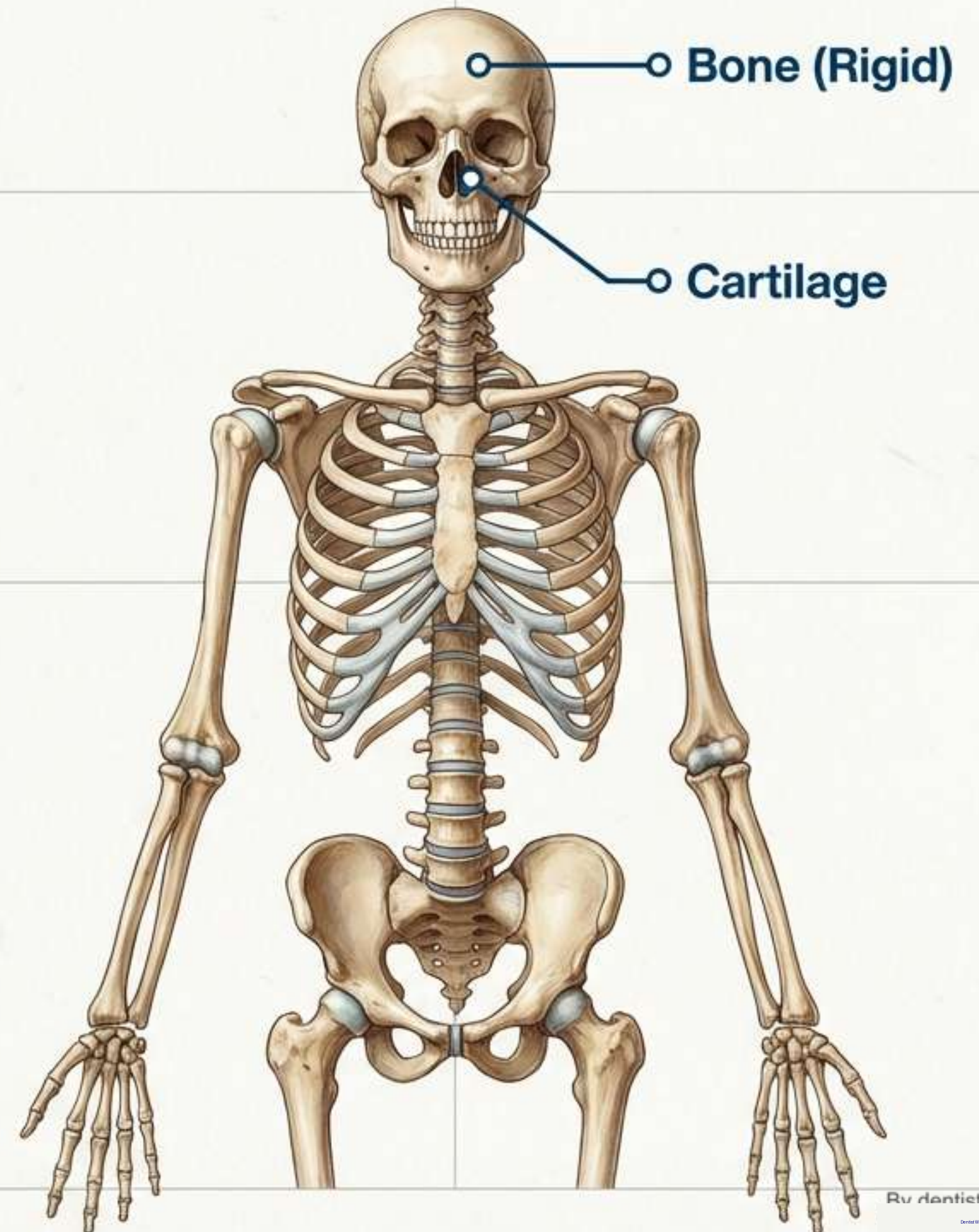
- Made up of: Bones and Cartilage.
- Osteology: The study of bones.
- Adult Skeleton: Made up of 206 bones and weighs around 20% of the body's weight [Ref: Q6 A].
- Supernumerary bones: May exist (inconstant bones).

DEFINITION: BONE

A highly specialized and rigid connective tissue [Ref: Q1 A, Q6 C] formed by cells called osteocytes.

Examples: Bones of the skull, bones of the limbs.

Note: Bone is a 'Phosphocalcique' reservoir [Ref: Q1 E].



2. DEFINITIONS: CARTILAGE & TYPES

CARTILAGE:

An elastic and semi-rigid connective tissue formed by cells called chondrocytes.

Characteristic: It has no nerves or blood vessels.

1. HYALINE CARTILAGE

- Formed by spherical chondrocytes, rich in collagen fibers.
- Most common type.
- Locations: The joints (articular) [Ref: Q4 B]
Rib (costal)
Larynx
Trachea/Bronchi
Nose.

2. ELASTIC CARTILAGE

- Contains elastic fibers.
- Locations: Ear and epiglottis.

3. FIBROUS CARTILAGE

- Thick collagen fiber bundles + alternating chondrocytes.
- Locations (Compression areas):
Intervertebral discs
Knee menisci
Labrums.

Elastic

Contains elastic fibers, providing flexibility

Fibrous

Contains thick collagen fibers, supporting shear and compressive forces

Hyaline

Flexible but cartilage of the articular cartilage of the knee, and alting the articular cartilage and impact



3. FUNCTIONAL ANATOMY & 4. STRUCTURE

FUNCTIONAL ANATOMY OF CARTILAGE

The essential role is mechanical.

- Protects joint surfaces from wear and tear.
- Provides attachment for muscles.
- Allows passages (e.g., trachea) to remain permanently open.

STRUCTURE OF THE HUMAN SKELETON

Consists of an Axial skeleton to which the **Appendicular skeleton** is attached [Ref: Q11 A, Q12 A, Q13 E, Q14 A].

A. THE AXIAL SKELETON

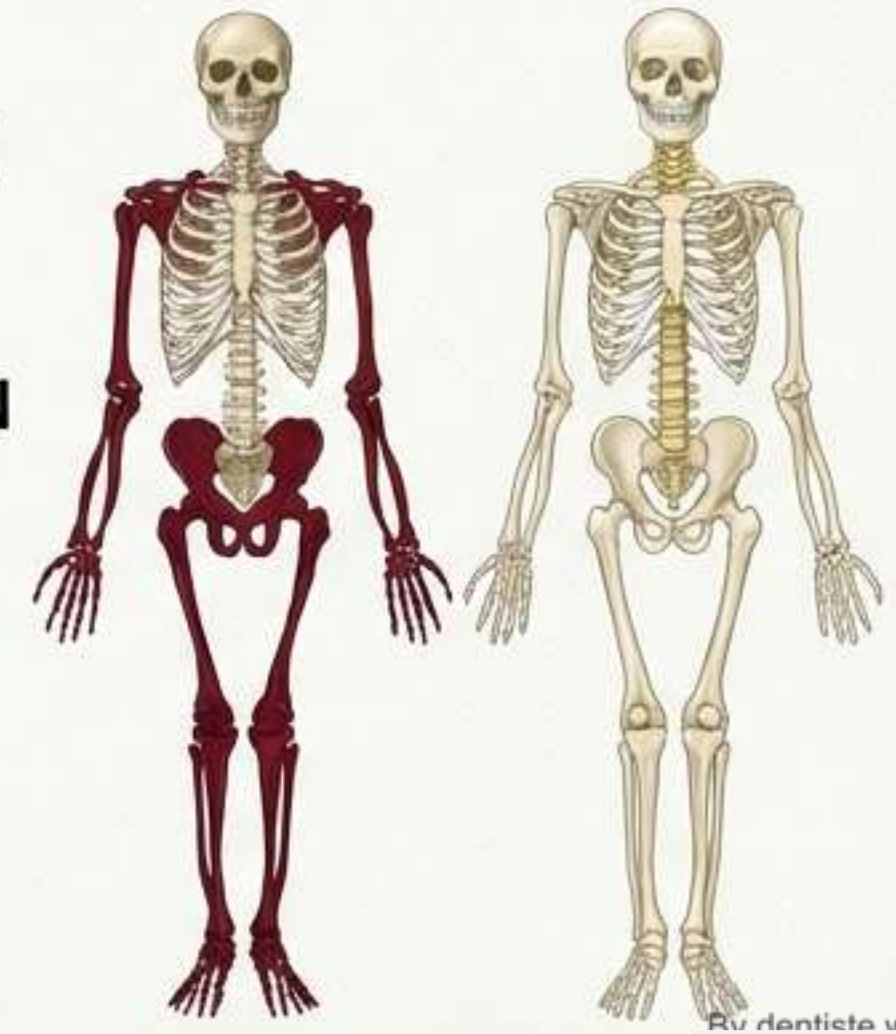
Forms the body's axis and supports it. Includes:

- Skull & Hyoid bone
- **Spine (Vertebral Column)** [Ref: Q9]
- **Ribs and Sternum** [Ref: Q9]

B. THE APPENDICULAR SKELETON

Corresponds to the limbs ("appendages").

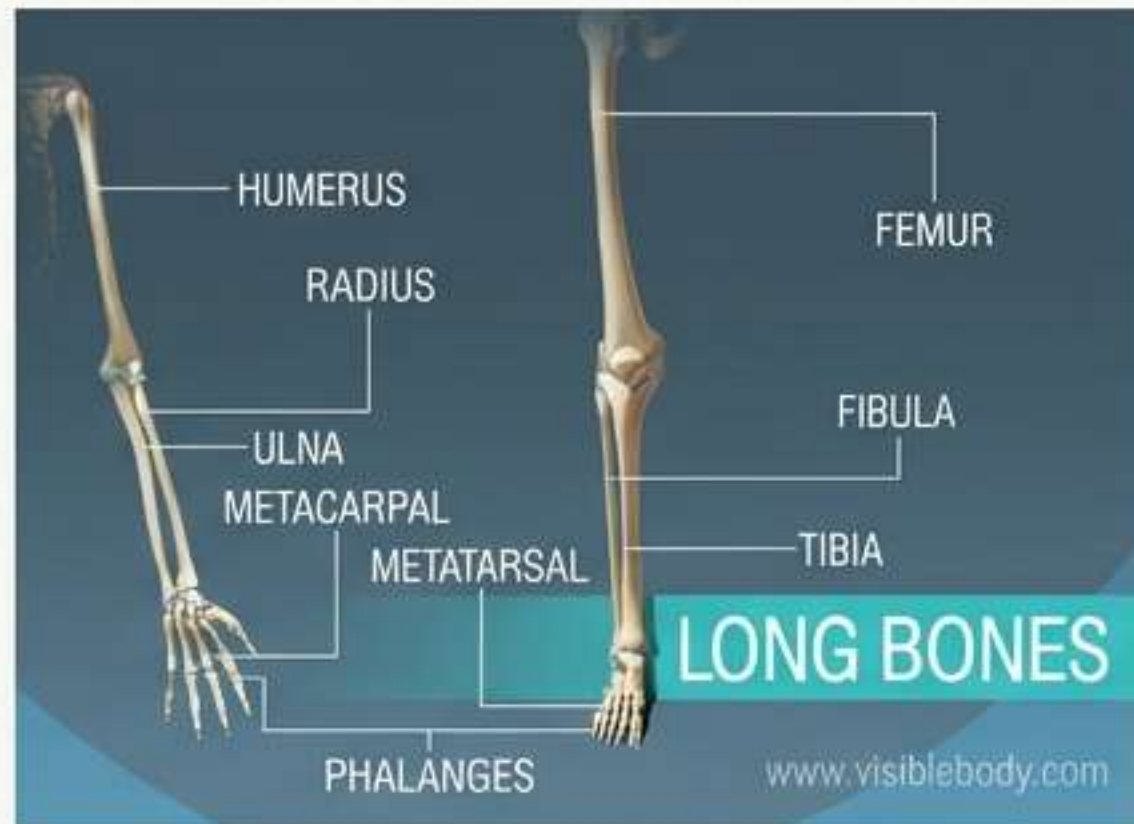
- Attaches to axial skeleton via the **shoulder girdle and pelvic girdle** [Ref: Q16 B].



5. CLASSIFICATION OF BONES (PART 1)

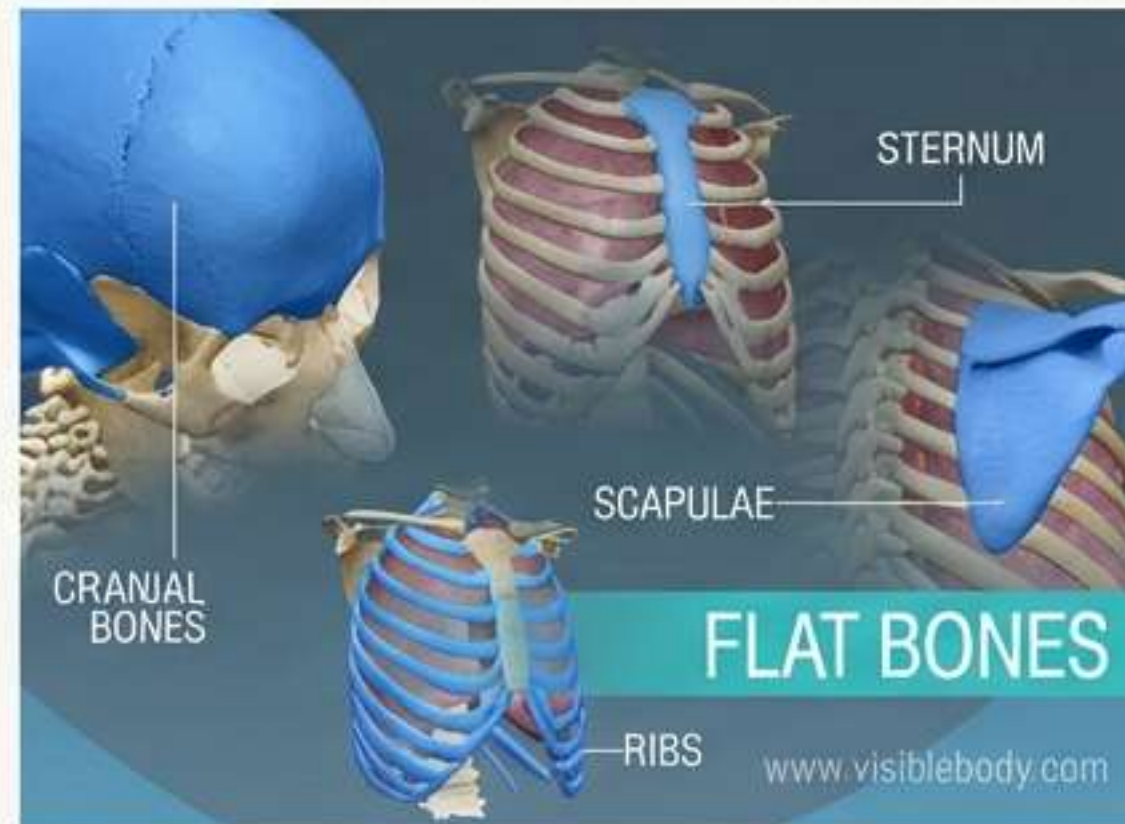
LONG BONES

- Length is greater than width and thickness [Ref: Q3 A, Q8 A, Q10 A, Q15 B, Q17 B].
- Have a shaft (diaphysis) and two ends (epiphyses) [Ref: Q3, Q13 D, Q14 D].
- Example: The Humerus [Ref: Q15 E, Q17 E].



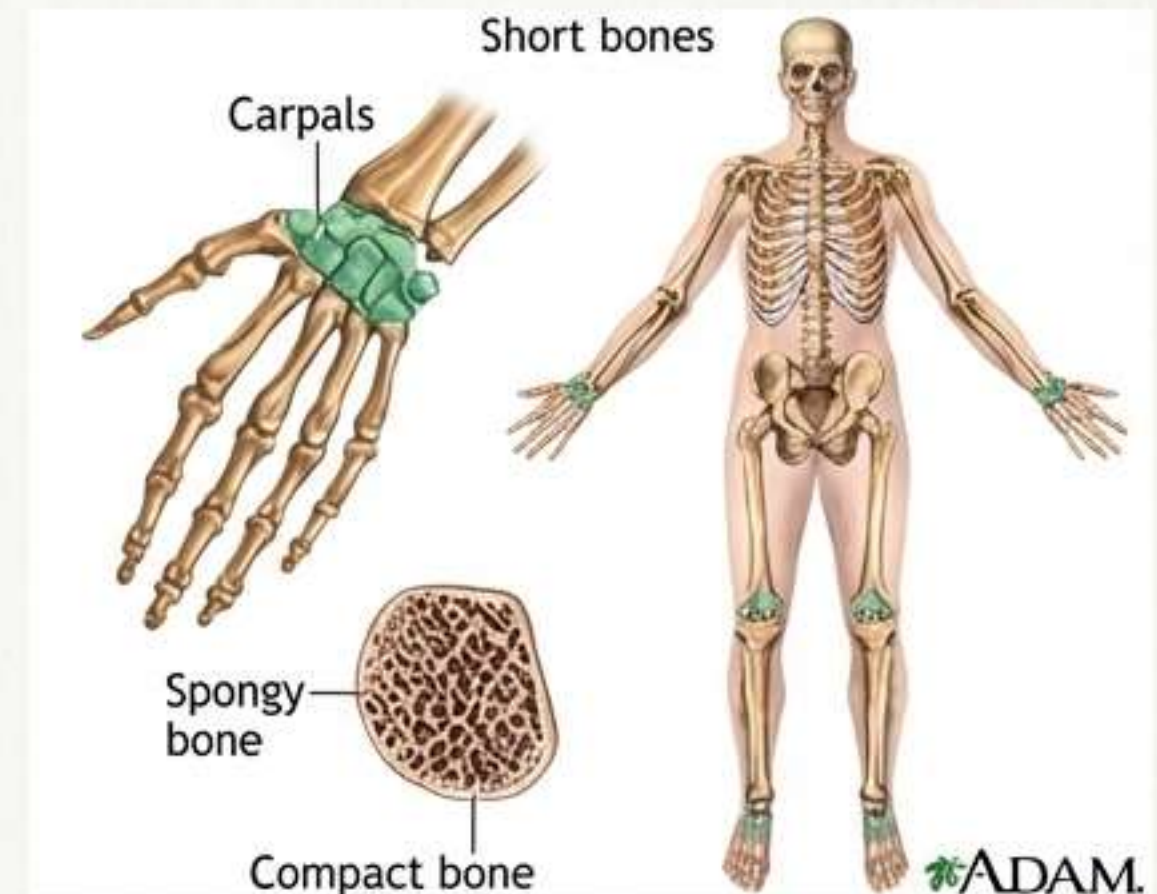
FLAT BONES

- Length and width are more important than thickness.
- Example: The Scapula (shoulder bone) [Ref: Q12 B].
- Other Examples: Sternum, Cranial bones.



SHORT BONES

- All three dimensions are equal.
- Example: The Carpal bones of the wrist [Ref: Q12 C].



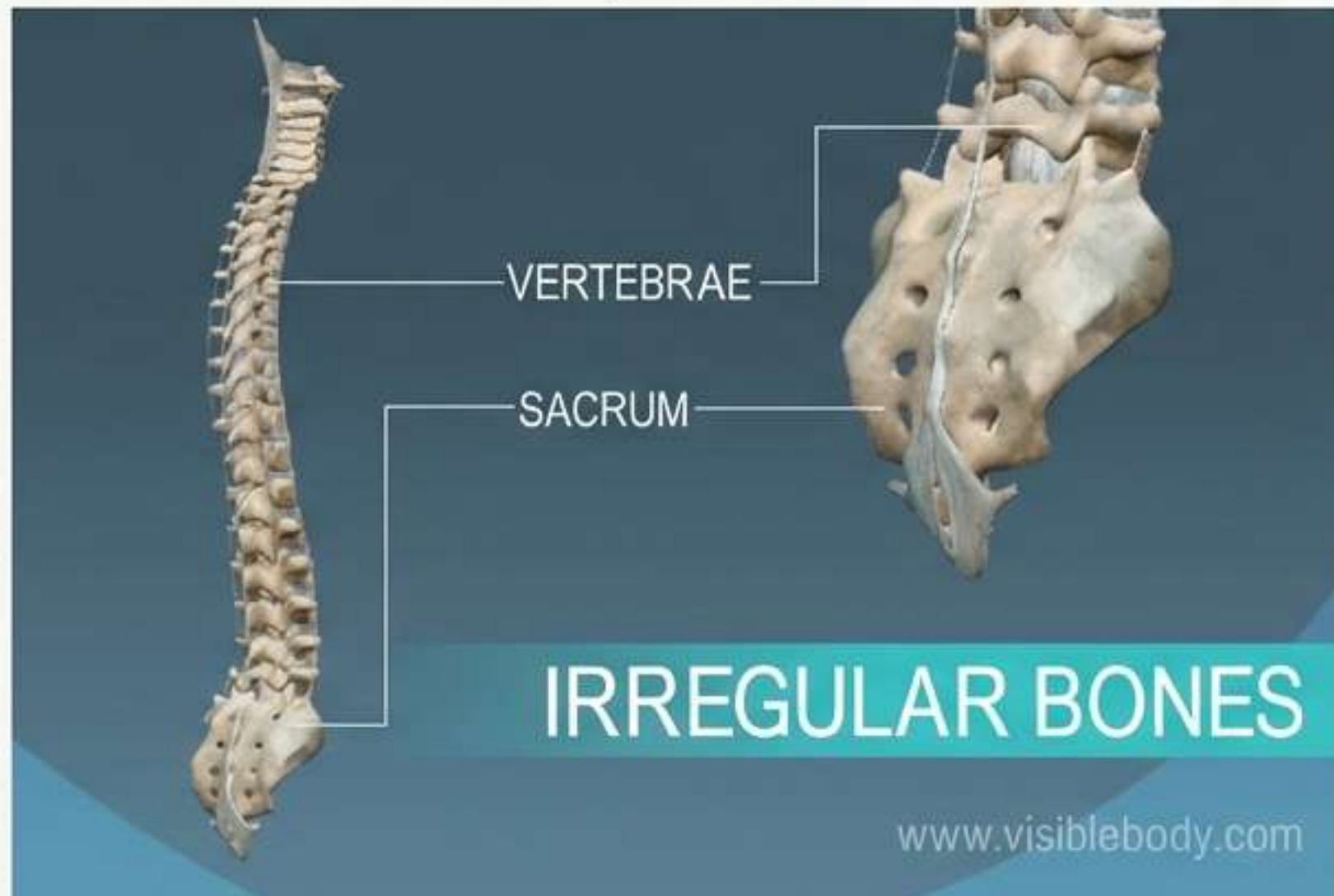
5. CLASSIFICATION (PART 2) & 6. MORPHOLOGY

IRREGULAR BONES

- Unclassifiable shapes.
- Example: Vertebrae, Maxillary bone.

SPECIAL TYPES

- **Sesamoid bones:** Embedded in tendons (e.g., Patella).
- **Vomer bones:** Variable, located between skull bones.



BONE MORPHOLOGY

Surface features: Protrusions (eminences), Depressions (cavities), Foramina (holes).

A. PROTRUSION (ELEVATION)

1. Articular:

- Head
- **Condyle** [Ref: Q4 C]

2. **Non-articular** (Insertion for ligaments/muscles):

- Processes, Tuberosities, Tubercles, Spines, Crests.

6. MORPHOLOGY (CONT.) & 7. BONE TISSUE

B. DEPRESSION (CAVITY)

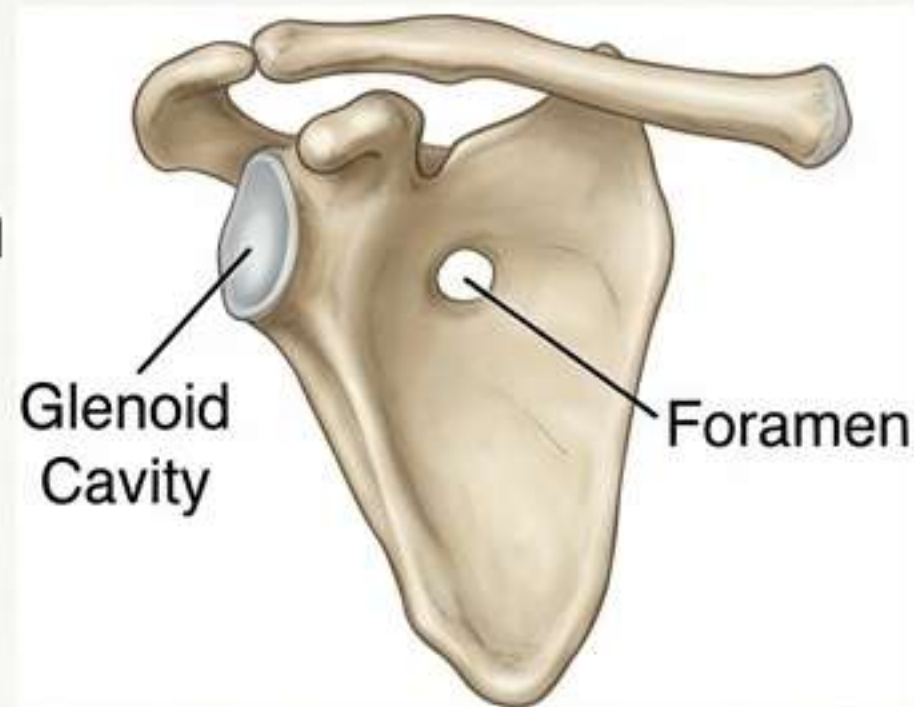
- Articular: Forms a joint (e.g., Glenoid cavity of scapula).
- Non-articular: Muscle insertion or vessel passage (Dimples, Furrows).

C. FORAMEN

- Opening for nerves or vessels.

D. INCISURES

- Indentations on the edge.



BONE TISSUE: THE CELLS

1. **Osteoblasts:** Form bone tissue; secrete collagen necessary for ossification. [Ref: Q7 C]
2. **Osteoclasts:** Cells that **destroy bone** (resorption) while osteoblasts rebuild it [Ref: Q7 C].
3. **Osteocytes:** Come from osteoblasts; maintain calcium-phosphorus balance.
4. **Border cells:** Located on surface.

7. INTERNAL STRUCTURE: THE PERIOSTEUM

1. THE PERIOSTEUM

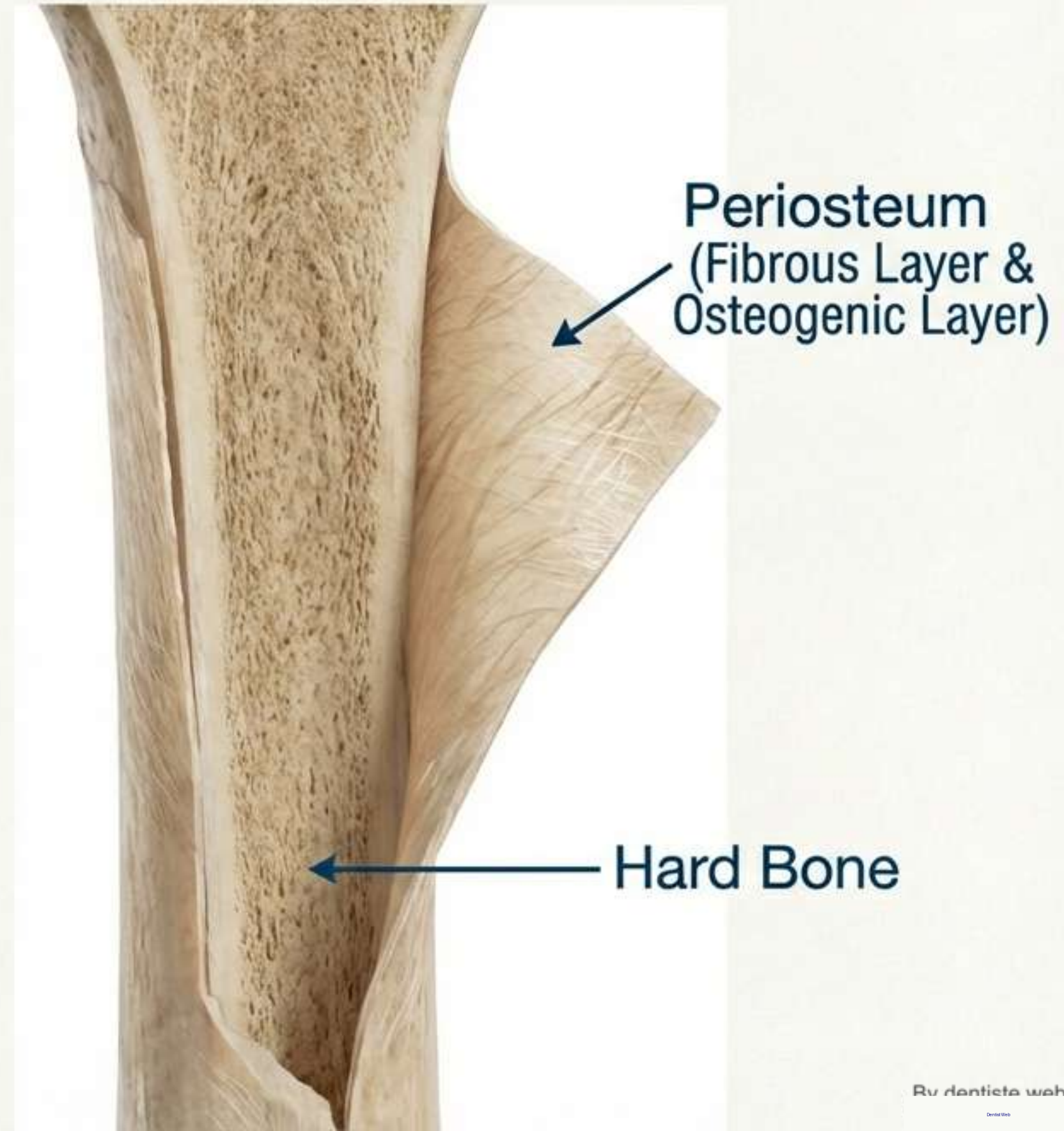
- Fibrous membrane that covers the entire surface of the bone [Ref: Q2 A, Q7 B].
- EXCEPTION: Does NOT cover cartilaginous (articular) surfaces [Ref: Q2 D].

COMPOSITION:

- Inner osteogenic layer.
- Outer fibrous layer.

FUNCTIONS:

- Allows muscles and tendons to attach.
- Richly vascularized and innervated.
- Ensures bone growth in width.
- Ensures bone consolidation (fracture repair).



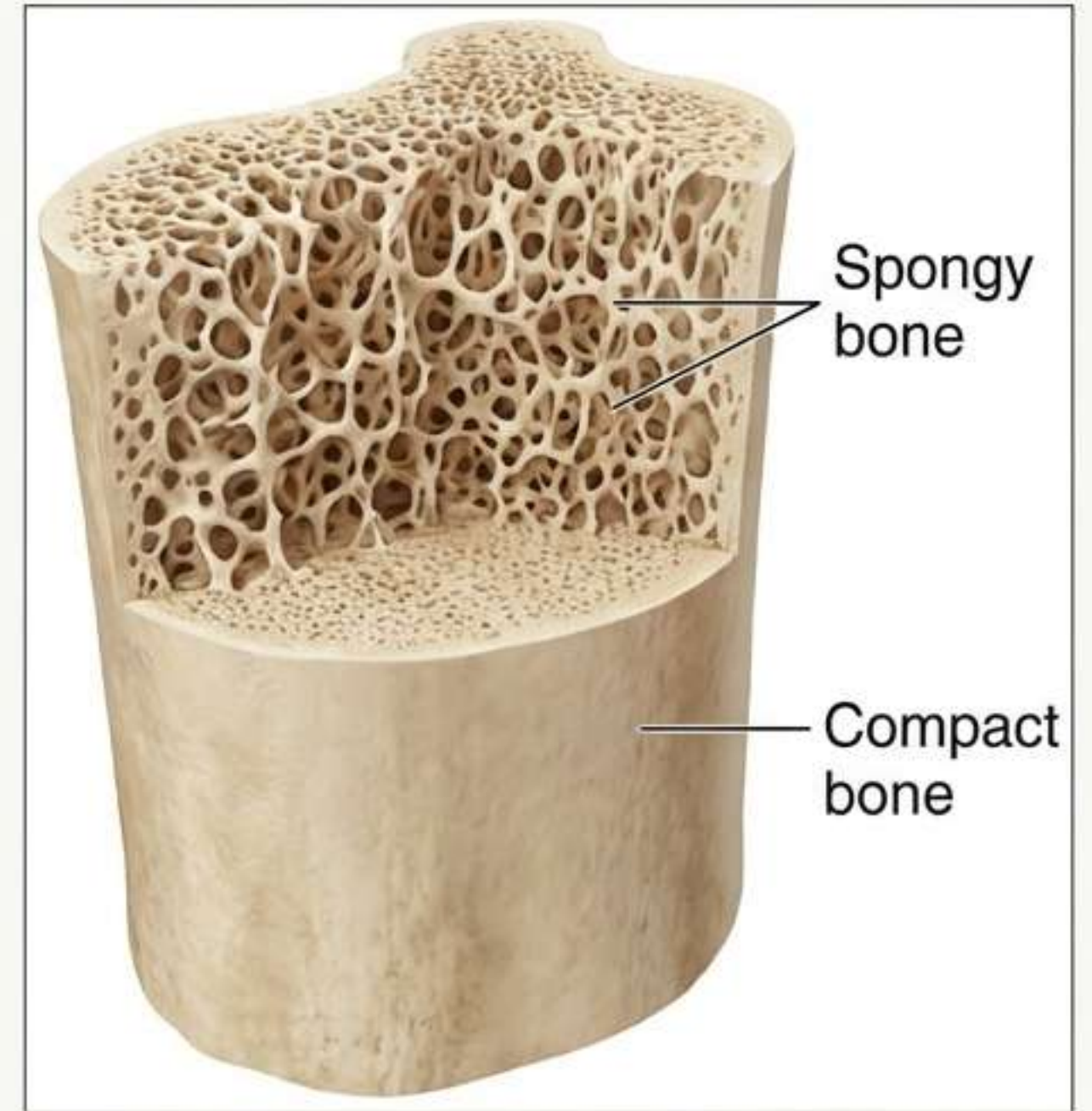
7. INTERNAL STRUCTURE: BONE TYPES

2. COMPACT BONE

- Very dense, uniform, and hard.
- Formed by Osteons (Haversian system) and bone lamellae.
- Accounts for 80% of bone surface area.

3. SPONGY BONE (CANCELOUS)

- Brittle and hollowed out by bone lamellae.
- Location: Contained in the **Epiphyses of long bones** [Ref: Q2 C, Q2 D].
- Forms the intermediate layer of flat and short bones.



7. INTERNAL STRUCTURE: CAVITIES & MARROW

4. THE BONE CAVITY

- Empty/Air-filled: Paranasal sinuses.
- Filled with substance: Medullary canal.
- Endosteum: Osteogenic membrane covering cavities (fracture consolidation role).

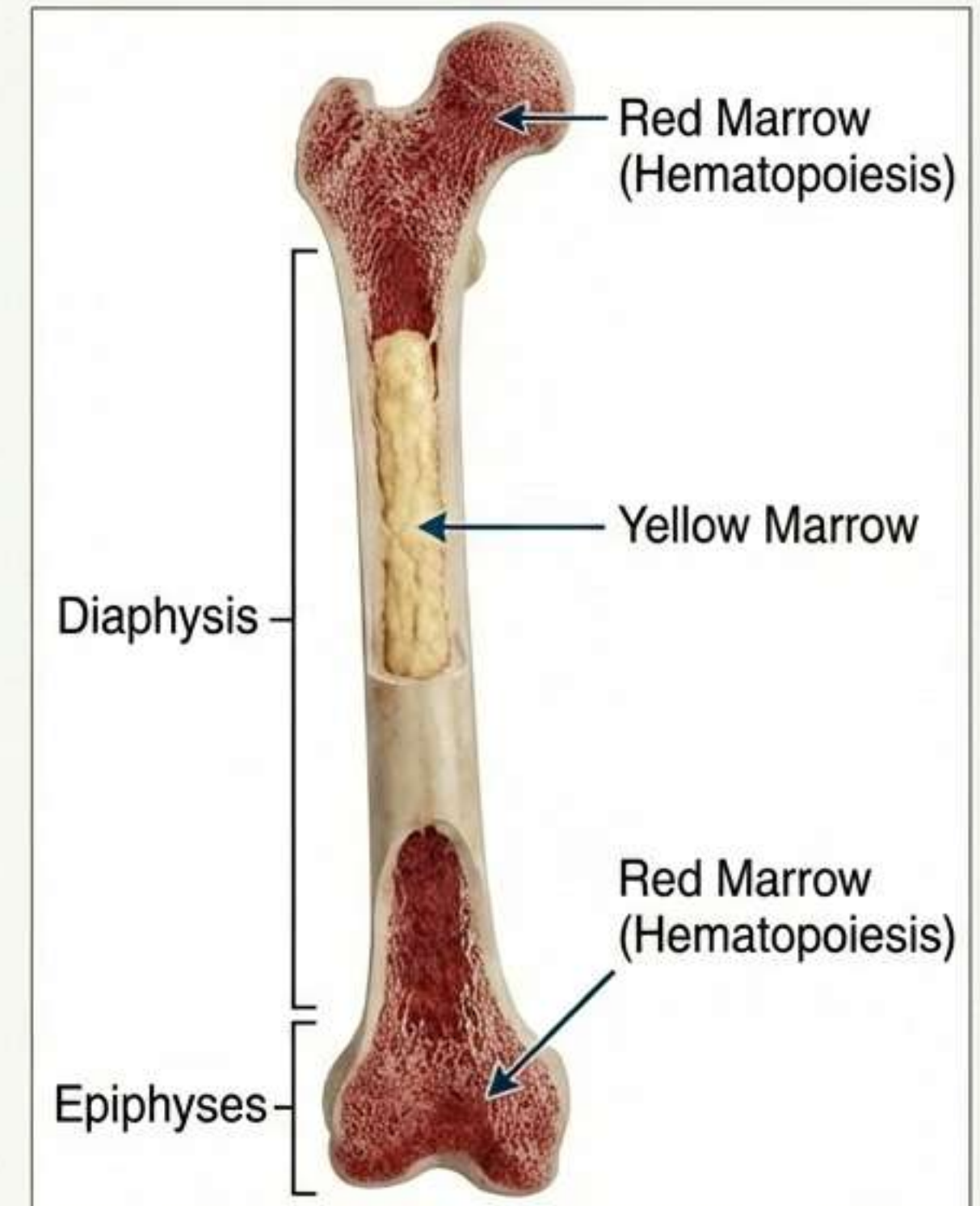
BONE MARROW

Yellow Marrow:

- Fatty mass.
- Occupies the center of the diaphysis in adults [Ref: Q7 A].

Red Marrow (Hematopoiesis):

- Produces blood cells (100-150 billion RBCs/day).
- Occupies spongy bone.
- *Note: Found in diaphysis in children, but restricted to spongy bone in adults.*



7. EXTERNAL STRUCTURE: LONG BONES

1. DIAPHYSIS (SHAFT)

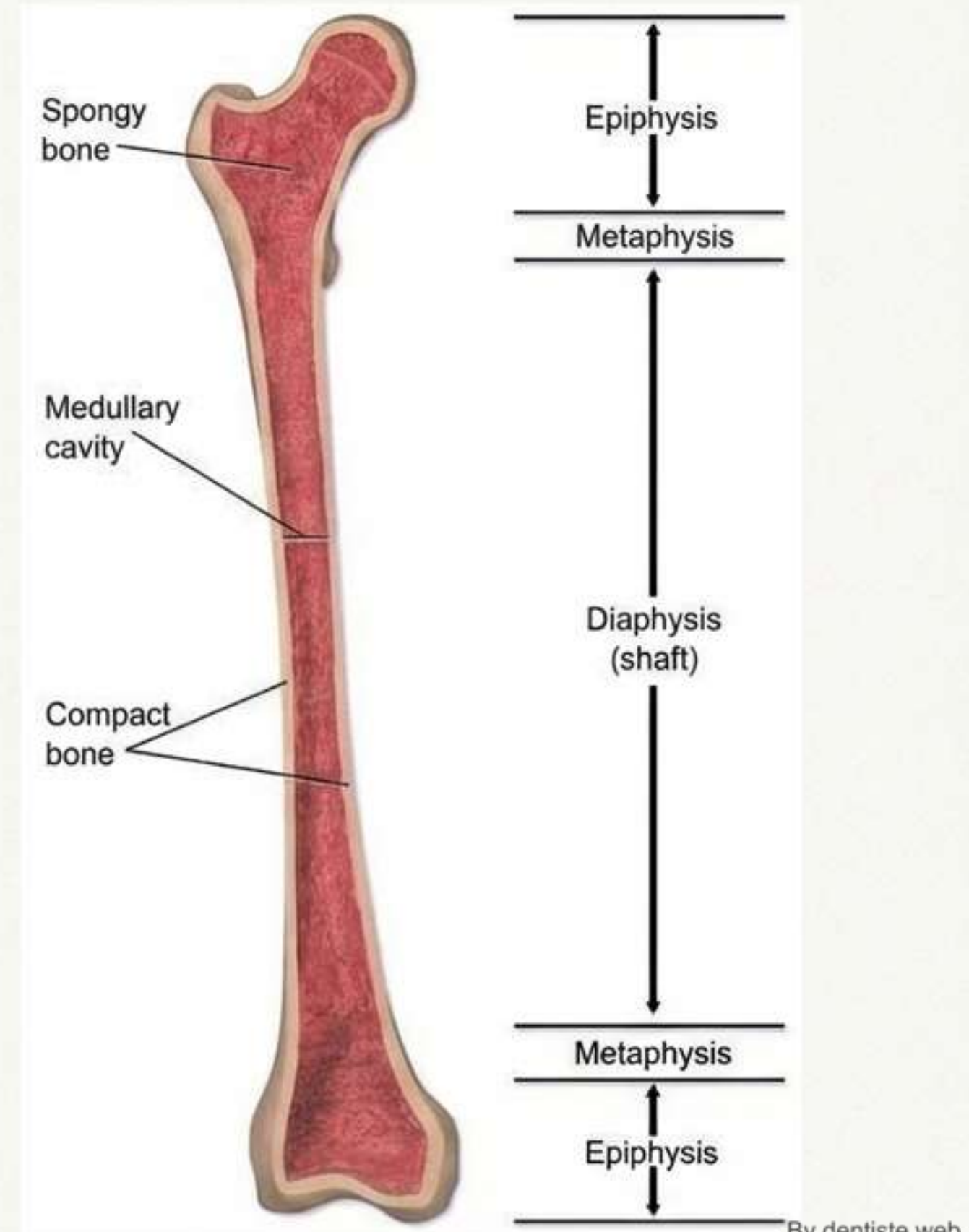
- Compact bone hollowed out by the **central cavity** [Ref: Q7 A].
- Contains red marrow (child) or yellow marrow (adult).

2. EPIPHYSES (ENDS)

- Located at ends of diaphysis.
- Formed of **spongy tissue** [Ref: Q2 D].
- Covered by **cartilaginous articular surfaces** [Ref: Q4 B, Q8 E].

3. METAPHYSIS

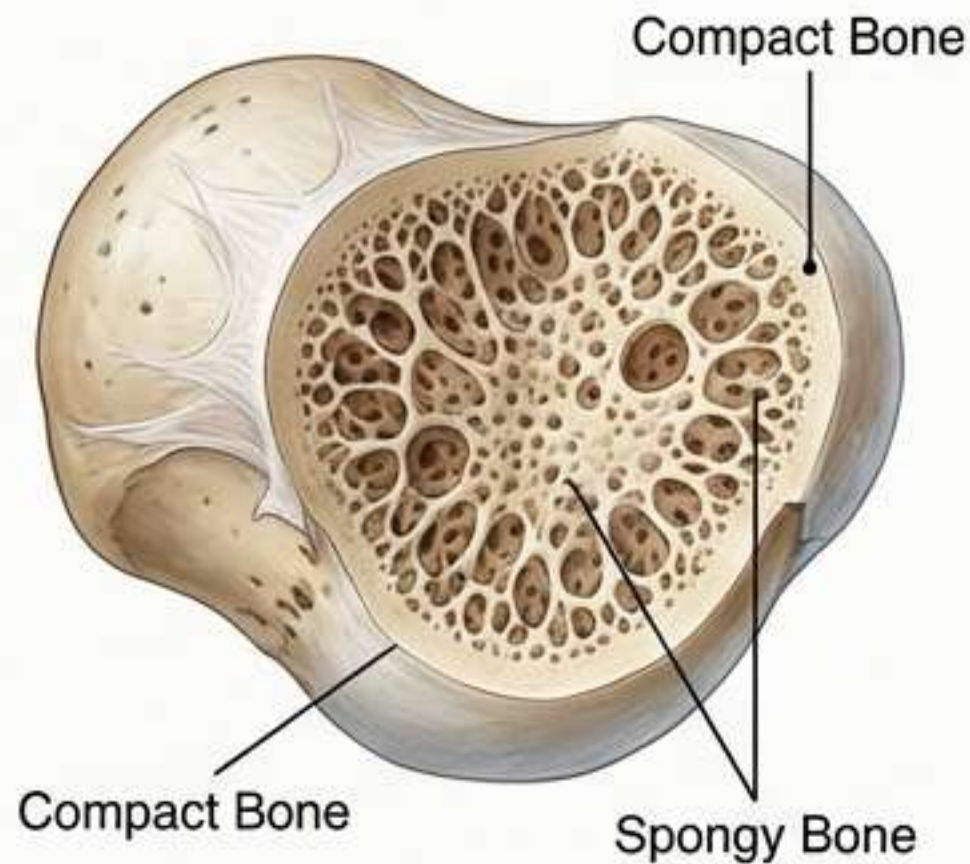
- Segment between epiphysis and diaphysis.
- Contains **growth cartilage** in children [Ref: Q7 D, Q7 E].



7. EXTERNAL STRUCTURE: SHORT, FLAT & IRREGULAR

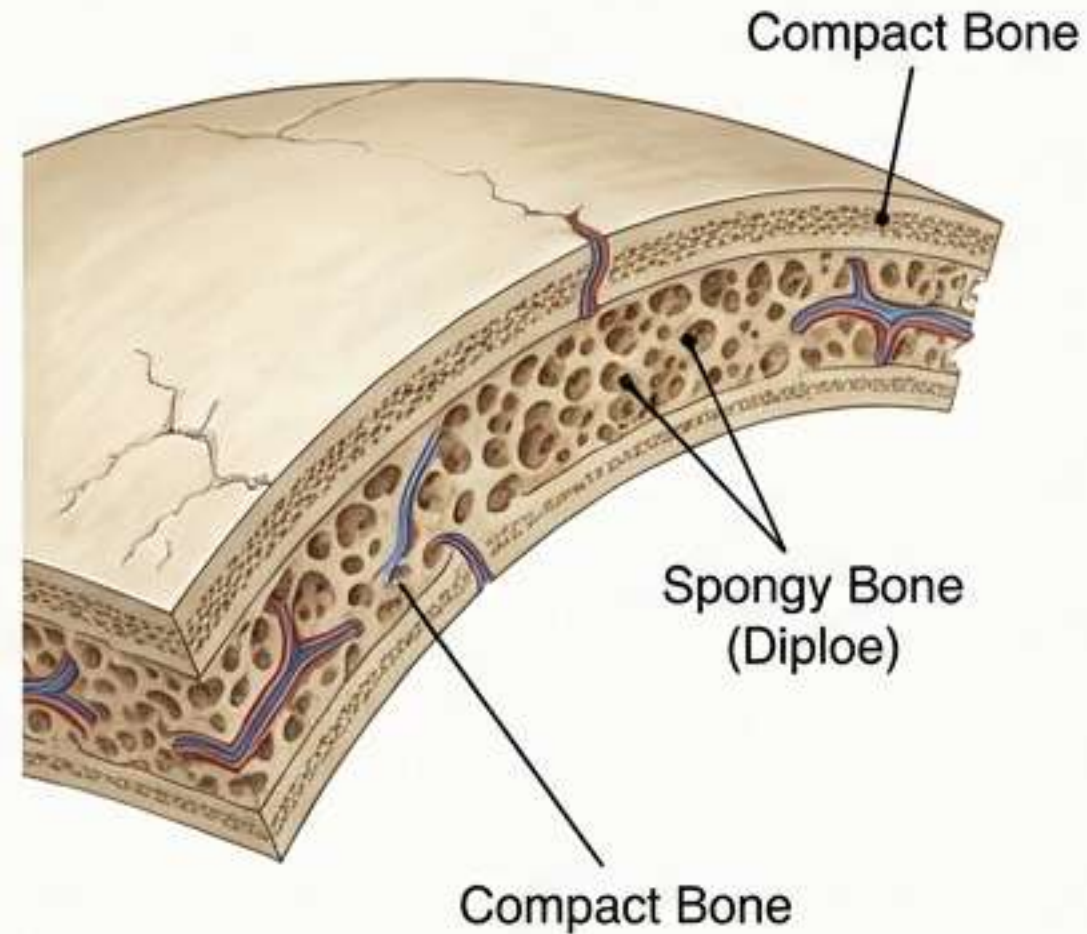
2. SHORT BONES

Structure: A layer of spongy substance and marrow contained within a thin layer of compact substance. Covered by periosteum.



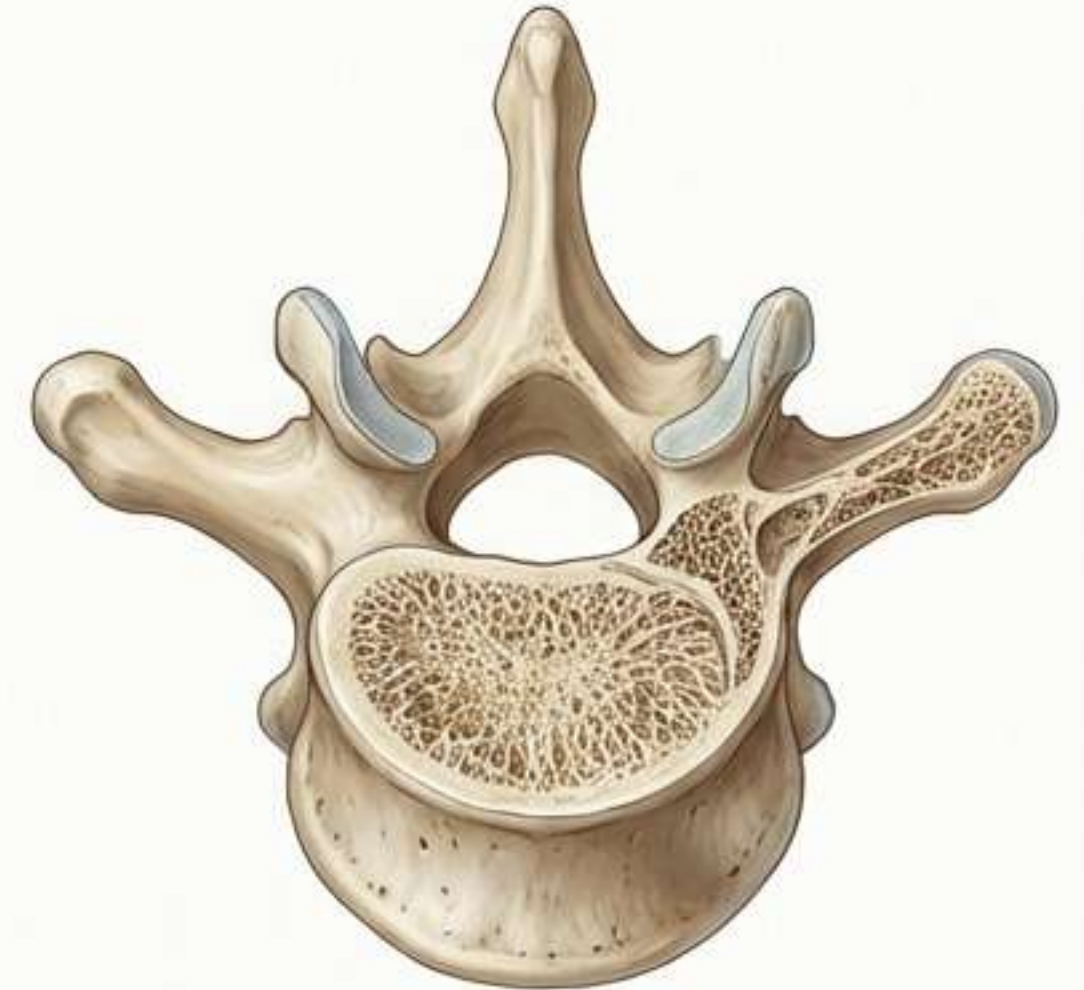
3. FLAT BONES

Structure: Two layers of compact bone surrounding a layer of spongy bone.



4. IRREGULAR BONES

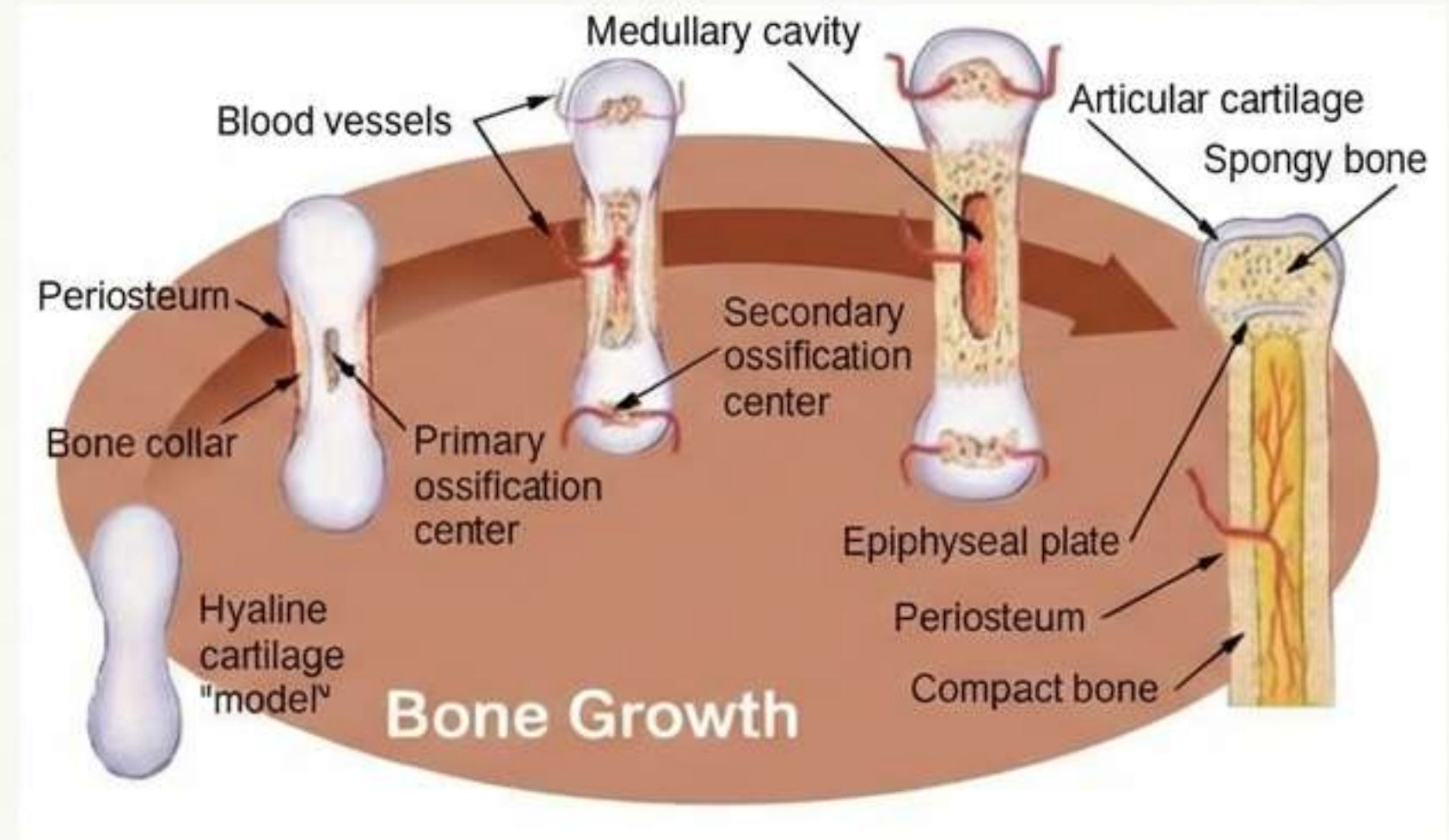
Structure: Combine these different structures.



8. DEVELOPMENT (OSTEOGENESIS) & 9. GROWTH

OSTEOGENESIS (Two Types)

1. **Endochondral (Cartilaginous):** Hyaline cartilage replaced by bone. (Long/Short bones).
2. **Fibrous (Membranous):** Connective tissue transforms directly to bone. (Flat bones/Skull).



BONE GROWTH (Stops age 25-28)

- **Growth in Width:** Ensured by the **Periosteum**.
- **Growth in Length:** Ensured by the **Growth Plate (cartilage)** located at the metaphysis [Ref: Q7 E, Q8 D, Q10 E].
- **Flat Bones:** Grow at bone **sutures**.

10. VASCULARIZATION & INNERVATION

LONG BONES ARTERIES

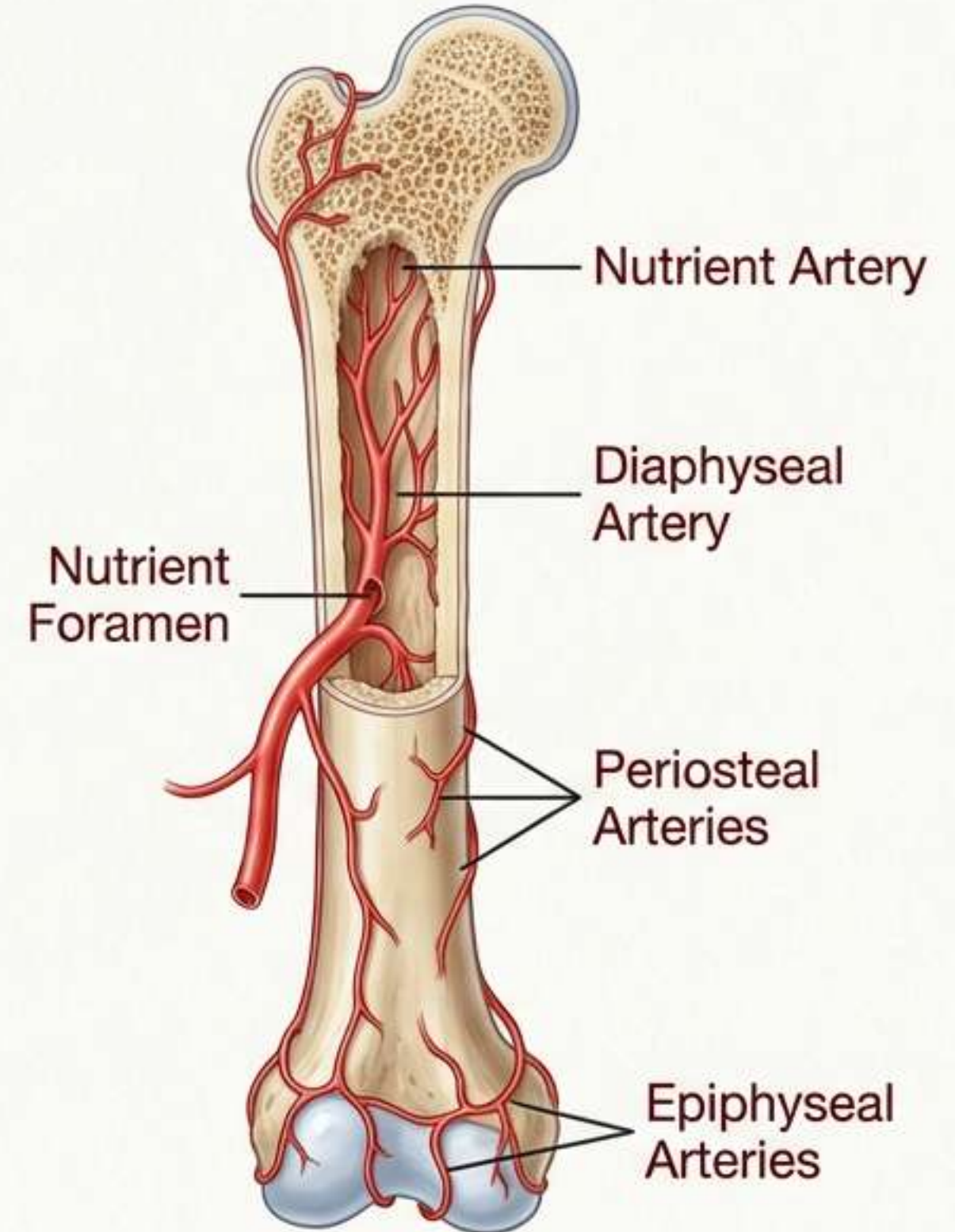
1. **Diaphyseal arteries:** Enter through the **nutrient foramen** [Ref: Q3 D].
2. **Epiphyseal arteries:** Form a periarticular circle.
3. **Periosteal arteries.**

FLAT & SHORT BONES

- **Flat:** Numerous arteries entering nutrient foramen.
- **Short:** Only periosteal arteries.

NERVES

- Satellite to arteries.
- Vasomotor and sensory.



11. FUNCTIONAL ANATOMY & PATHOLOGY

FUNCTIONAL ANATOMY

1. **Framework & Protection:** Protects organs [Ref: Q5 D].
2. **Mobility:** Enables movement thanks to joints [Ref: Q6 B].
3. **Mineral Reserve:** Particularly Calcium and Phosphorus [Ref: Q1 E, Q5 A].
4. **Hematopoiesis:** Production of blood cells in bone marrow [Ref: Q1 C, Q5 B].



BONE DISORDERS

- **Fractures:** Traumatic disorder.
- **Osteoporosis:** Resorption > Formation.
- **Osteomalacia:** Bone demineralization.
- **Osteoarthritis:** Cartilage disease.
- **Paget's disease:** Accelerated bone remodeling.